

Introduction: Network operating System VS Distributed operating systems, Characteristics, Design goals, challenges of Distributed Systems, Examples of Distributed Systems, Trends in Distributed systems: Pervasive networking and the modern Internet, Mobile and ubiquitous computing, Focus on resource sharing Distributed Computing Models: Physical, Architecture and Fundamental models Introduction to middleware, middleware Framework, Role of middleware, Examples of Middleware, Origins of middleware, Architecture vs Middleware, RMI, CORBA, General Approaches to adaptive software, Types of middleware-messages oriented middleware, intelligent middleware, content centric middleware, middleware protocol, middleware Services, Distributed computing Environment (DCE), middleware Issues, middleware Analyst IPC: Introduction, Layered protocols, API for internet protocols, IPC through shared memory, external data representation and marshaling, Types of communication, inter process communication, multicast communication, message-oriented communication, MPI, network virtualization, overlay networks Coordination: Clock synchronization, logical clocks, mutual exclusion, election algorithms, Gossip based coordination Replication: Reasons for replication, Replica management – Finding the best server location, Content replication and placement, Content distribution, Managing replicated objects Consistency protocols: Primary based protocols, replicated write protocols Fault Tolerance: Introduction to fault tolerance, Reliable client server communication, Reliable group communication, distributed commit, Recovery – Check pointing, Message logging Distributed Files: Introduction, File System Architecture, Sun Network File System and HDFS. Distributed Multimedia Systems: Characteristics of Multimedia Data, Quality of Service Management, Resource Management Distributed Web Based Systems: Architecture of Traditional Web-Based Systems, Apache Web Server, Web Server Clusters, Communication by Hypertext Transfer Protocol, Synchronization, Web Proxy Caching Recent Trends: Introduction, Portable and handheld Devices, Wearable devices, Devices embedded in appliances, Parallel Virtual Machine (PVM), Jini, Service Oriented Architecture, The Future of Recent Trends. Tools for Distributed System Monitoring: Prometheus, Zabbix, Nagios this are some topics of distributed system and i have viva tomorrow and give me questions of this topics with one to two lines for viva

Distributed System Viva Questions & Short Answers

Introduction to Distributed Systems

1. What is a Distributed System?

A distributed system is a collection of independent computers that work together as a single system.

2. Difference between Network OS and Distributed OS?

Network OS allows resource sharing between separate machines, while Distributed OS makes multiple machines appear as one system.

3. What are the characteristics of Distributed Systems?

Resource sharing, scalability, concurrency, fault tolerance, and transparency.

4. What are the design goals of Distributed Systems?

Transparency, openness, reliability, scalability, and performance.

5. What are the challenges in Distributed Systems?

Synchronization, security, fault handling, scalability, and network latency.

6. Give examples of Distributed Systems.

Cloud computing, Google search engine, ATM networks, distributed databases.

7. What is pervasive networking?

It means devices are connected everywhere through the Internet.

8. What is mobile and ubiquitous computing?

Computing services available anytime and anywhere using mobile devices.

9. What is resource sharing?

Sharing hardware, software, and data among multiple systems.

Distributed Computing Models

10. What is a physical model?

It describes the hardware organization of distributed systems.

11. What is an architectural model?

It defines how software components interact with each other.

12. What is a fundamental model?

It explains interaction, failure, and security aspects of systems.

Middleware

13. What is middleware?

Middleware is software that connects applications and distributed components.

14. What is the role of middleware?

It provides communication, coordination, and interoperability between systems.

15. Give examples of middleware.

CORBA, RMI, DCOM, Web Services.

16. What is RMI?

Remote Method Invocation allows Java objects to communicate remotely.

17. What is CORBA?

Common Object Request Broker Architecture enables communication between applications written in different languages.

18. Difference between architecture and middleware?

Architecture defines structure; middleware provides communication support.

19. What is Message-Oriented Middleware?

Middleware that exchanges messages between applications asynchronously.

20. What is intelligent middleware?

Middleware capable of adaptive decision-making and automation.

21. What is content-centric middleware?

Middleware focused on data/content instead of host locations.

22. What is DCE?

Distributed Computing Environment is middleware for distributed applications.

23. What are middleware issues?

Security, interoperability, scalability, and performance.

IPC (Inter Process Communication)

24. What is IPC?

IPC is communication between processes running on same or different systems.

25. What are layered protocols?

Protocols organized into layers where each layer performs specific tasks.

26. What is shared memory IPC?

Processes communicate using a common memory area.

27. What is marshaling?

Converting data into a transferable format.

28. Types of communication in distributed systems?

Unicast, multicast, and broadcast communication.

29. What is multicast communication?

Sending data to multiple receivers simultaneously.

30. What is message-oriented communication?

Communication using message passing between processes.

31. What is MPI?

Message Passing Interface used in parallel computing.

32. What is network virtualization?

Creating virtual versions of network resources.

33. What are overlay networks?

Virtual networks built on top of existing networks.

Coordination

34. What is clock synchronization?

Keeping clocks of distributed systems consistent.

35. What are logical clocks?

Clocks used to order events logically instead of real time.

36. What is mutual exclusion?

Ensuring only one process accesses a critical resource at a time.

37. What is an election algorithm?

Algorithm used to select a coordinator process.

38. What is gossip-based coordination?

Nodes exchange information periodically like gossip spreading.

Replication

39. Why is replication used?

For fault tolerance, high availability, and faster access.

40. What is replica management?

Managing copies of data across servers.

41. What is content replication?

Storing multiple copies of content at different locations.

42. What is content distribution?

Delivering content efficiently to users.

43. What are replicated objects?

Multiple copies of objects stored on different machines.

Consistency Protocols

44. What is consistency in distributed systems?

Ensuring all replicas have the same data.

45. What are primary-based protocols?

One primary server handles updates before replicas synchronize.

46. What are replicated-write protocols?

Updates are written to multiple replicas simultaneously.

Fault Tolerance

47. What is fault tolerance?

Ability of a system to continue working after failures.

48. What is reliable client-server communication?

Communication ensuring messages are delivered correctly.

49. What is reliable group communication?

Reliable communication among multiple distributed nodes.

50. What is distributed commit?

Ensuring all distributed transactions complete consistently.

51. What is checkpointing?

Saving system state periodically for recovery.

52. What is message logging?

Recording messages to recover from failures.

Distributed Files

53. What is a distributed file system?

A file system where files are stored across multiple machines.

54. What is file system architecture?

Structure defining file storage and access methods.

55. What is Sun NFS?

Network File System allowing remote file access.

56. What is HDFS?

Hadoop Distributed File System for big data storage.

Distributed Multimedia Systems

57. What are multimedia data characteristics?

Large size, real-time delivery, synchronization needs.

58. What is QoS management?

Managing bandwidth, delay, and reliability for multimedia.

59. What is resource management?

Efficient allocation of CPU, memory, and bandwidth.

Distributed Web-Based Systems

60. What is a traditional web-based architecture?

Client-server architecture using browsers and web servers.

61. What is Apache Web Server?

Popular open-source web server software.

62. What are web server clusters?

Multiple servers working together for load balancing.

63. What is HTTP?

HyperText Transfer Protocol used for web communication.

64. What is web proxy caching?

Storing web content temporarily to improve speed.

Recent Trends

65. What are portable devices?

Small mobile devices like smartphones and tablets.

66. What are wearable devices?

Devices worn on the body like smartwatches.

67. What are embedded devices?

Computing devices built into appliances and machines.

68. What is PVM?

Parallel Virtual Machine used for parallel processing.

69. What is Jini?

Java-based technology for distributed services.

70. What is SOA?

Service Oriented Architecture where services communicate over networks.

71. What is the future of distributed systems?

Cloud computing, IoT, AI integration, and edge computing.

Monitoring Tools

72. What is Prometheus?

Open-source monitoring tool mainly for cloud systems.

73. What is Zabbix?

Monitoring software for servers, networks, and applications.

74. What is Nagios?

Tool used for infrastructure and network monitoring.

75. Difference between Prometheus and Nagios?

Prometheus focuses on metrics collection; Nagios focuses on infrastructure monitoring.

Important Quick Viva Questions

76. What is transparency in distributed systems?

Hiding complexity from users.

77. What is scalability?

Ability to handle increasing workload efficiently.

78. What is concurrency?

Multiple processes executing simultaneously.

79. What is latency?

Delay in communication over a network.

80. What is load balancing?

Distributing workload among servers evenly.

1. Implement multi-threaded client/server Process communication using RMI. 2. Develop any distributed application using CORBA to demonstrate object brokering. (Calculator or String operations). 3. Develop a distributed system, to find sum of N elements in an array by distributing N/n elements to n number of processors MPI or OpenMP. Demonstrate by displaying the intermediate sums calculated at different processors. 4. Implement Berkeley algorithm for clock synchronization. 5. Implement token ring based mutual exclusion algorithm. 6. Implement Bully and Ring algorithm for leader election. these are some practices we have to perform give me question and answer on this

RMI Practical Viva

Multi-threaded Client/Server Communication using RMI

1. What is RMI?

Java RMI (Remote Method Invocation) allows methods of remote Java objects to be called from another JVM.

2. Why is RMI used in distributed systems?

It enables communication between distributed Java applications.

3. What are the main components of RMI?

Client, Server, Remote Interface, Stub, and RMI Registry.

4. What is a remote interface?

An interface that declares methods accessible remotely.

5. Why do RMI methods throw RemoteException?

To handle network-related communication errors.

6. What is the role of RMI Registry?

It stores remote object references for clients to access.

7. What is a stub in RMI?

A stub acts as a proxy between client and remote object.

8. What is multithreading in RMI server?

Handling multiple client requests simultaneously using threads.

9. Advantage of multi-threaded server?

Improves performance and supports concurrent clients.

10. Difference between local and remote method calls?

Remote calls involve network communication; local calls do not.

CORBA Practical Viva

Distributed Calculator/String Operations using CORBA

11. What is CORBA?

CORBA stands for Common Object Request Broker Architecture.

12. What is object brokering?

Connecting client requests to remote objects transparently.

13. What is ORB in CORBA?

Object Request Broker handles communication between distributed objects.

14. What is IDL?

Interface Definition Language used to define CORBA interfaces.

15. Why is CORBA platform-independent?

Because it supports multiple programming languages and systems.

16. What is a servant in CORBA?

Server-side object implementing remote methods.

17. What is naming service in CORBA?

Service used to locate remote objects.

18. Advantages of CORBA?

Language independence, interoperability, scalability.

19. Difference between RMI and CORBA?

RMI is Java-specific; CORBA supports multiple languages.

20. Why is CORBA called middleware?

Because it connects distributed applications.

MPI / OpenMP Practical Viva

Sum of N Elements using Distributed Processing

21. What is MPI?

Message Passing Interface used for parallel and distributed computing.

22. What is OpenMP?

API for shared-memory parallel programming.

23. Difference between MPI and OpenMP?

MPI uses distributed memory; OpenMP uses shared memory.

24. Why distribute array elements among processors?

To reduce computation time using parallelism.

25. What is parallel processing?

Executing multiple computations simultaneously.

26. What is a processor/node in distributed systems?

A computing unit performing assigned tasks.

27. What is intermediate sum?

Partial sum calculated by each processor.

28. What is speedup in parallel computing?

Performance improvement achieved using multiple processors.

29. What is synchronization in MPI/OpenMP?

Coordinating processes or threads during execution.

30. Advantage of distributed computation?

Faster execution and better resource utilization.

Berkeley Algorithm Viva

Clock Synchronization

31. What is clock synchronization?

Keeping all system clocks consistent in distributed systems.

32. What is Berkeley Algorithm?

Algorithm where a master synchronizes clocks of all nodes.

33. Does Berkeley algorithm use UTC time?

No, it uses average system time.

34. What is the role of master node?

Collects clock times and calculates average time.

35. Why is synchronization important?

To maintain event ordering and consistency.

36. How is average time calculated?

By averaging times from all participating nodes.

37. What happens if a clock differs too much?

It may be ignored as faulty.

38. Advantage of Berkeley algorithm?

Works well in systems without external time source.

39. Limitation of Berkeley algorithm?

Master node failure affects synchronization.

40. Difference between Berkeley and Cristian algorithm?

Cristian uses time server; Berkeley uses average of clocks.

Token Ring Mutual Exclusion Viva

Token Ring Algorithm

41. What is mutual exclusion?

Ensuring only one process accesses critical section at a time.

42. What is Token Ring algorithm?

Processes arranged in a ring pass a token for resource access.

43. What is a token?

Special permission message allowing critical section access.

44. Why is token needed?

To prevent simultaneous resource access.

45. What is critical section?

Part of program accessing shared resources.

46. Advantage of Token Ring algorithm?

No starvation and orderly access.

47. What happens if token is lost?

System must generate a new token.

48. What is starvation?

A process waiting indefinitely for resource access.

49. Is Token Ring centralized?

No, it is a decentralized algorithm.

50. Limitation of Token Ring?

Failure of one node may affect the ring.

Leader Election Algorithms Viva

Bully Algorithm

51. What is leader election?

Selecting one coordinator process among distributed nodes.

52. What is Bully Algorithm?

Highest-ID active process becomes coordinator.

53. Why is it called Bully algorithm?

Higher priority process dominates lower ones.

54. What happens when coordinator fails?

Election process starts automatically.

55. Advantage of Bully algorithm?

Simple and fast for small systems.

56. Limitation of Bully algorithm?

High message overhead.

Ring Algorithm

57. What is Ring Election Algorithm?

Processes arranged in ring pass election messages.

58. How is leader selected in Ring algorithm?

Process with highest ID becomes coordinator.

59. Difference between Bully and Ring algorithm?

Bully uses broadcasting; Ring uses circular message passing.

60. Advantage of Ring algorithm?

Lower network traffic compared to Bully.

61. What is coordinator process?

Process managing coordination tasks.

62. What is process ID in election algorithms?

Unique identifier assigned to each process.

Important Quick Viva Questions

63. What is distributed computing?

Multiple computers working together on tasks.

64. What is concurrency?

Multiple processes executing simultaneously.

65. What is fault tolerance?

System continues working after failures.

66. What is scalability?

Ability to handle increasing workload.

67. What is deadlock?

Processes waiting indefinitely for each other.

68. What is synchronization?

Coordinating process execution order.

69. What is latency?

Communication delay in networks.

70. Why are distributed systems important?

They improve performance, reliability, and resource sharing.